

Figure 5.1 Matvei Bronštejn.

The first to realize that our conceptual basis must change in order to understand quantum gravity was a romantic and legendary character: Matvei Bronštejn, a young Russian who lived during the Stalin era and died tragically.

Matvei

Matvei was a younger friend of Lev Landau – the scientist who would go on to become the best theoretical physicist of the Soviet Union. Colleagues who knew them both would claim that, of the two, Matvei was the more brilliant. At the point when Heisenberg and Dirac were constructing the bases of quantum mechanics, Landau, wrongly, thought that fields became ill defined due to quanta: quantum fluctuation would prevent us from measuring the value of a component of a field at a point (an arbitrary small region) in space. The masterly Bohr immediately saw that Landau was wrong, studied the issue in depth and wrote a long and detailed article to show that fields, such as the electric one, remain well defined even when quantum mechanics is brought to bear.[1] Landau dropped the issue.

But his young friend Matvei was intrigued, realizing that Landau's intuition, though imprecise, contained something of importance. He repeated the same reasoning by which Bohr had demonstrated that the quantum electric field was well defined at a point of space, applying it instead to the gravitational field, for which Einstein had just a few years previously written